



Internship Task Week 2

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Objective

The objective of this assignment was to gain practical experience with two fundamental cybersecurity tools:

1. A Python-based Password Strength Checker.
2. Nmap, a network scanning tool used to identify open ports on a system.

The task also involved understanding how weak passwords and open network ports can create security risks.

Part 1 – Password Strength Checker

Description

A Python program was created to evaluate the strength of passwords based on several security criteria.

The program checks whether the password:

- Contains at least 8 characters.
- Includes uppercase letters.
- Includes lowercase letters.
- Contains numbers.
- Contains special characters.
- Is not one of the commonly used passwords.

The program then categorizes the password as:

- Very Weak
- Weak
- Medium
- Strong

It also provides suggestions for improving weak passwords.

Password Evaluation Logic

Security Check

Minimum length (8 characters)

Uppercase letters

Purpose

Makes passwords harder to brute-force

Increases complexity

Security Check

Lowercase letters

Numbers

Special characters

Common password list

Purpose

Adds character variety

Improves password strength

Makes passwords more difficult to guess

Detects passwords that are frequently targeted

Python Code :

```
import string
```

```
common_passwords = [  
    "123456",  
    "password",  
    "qwerty",  
    "12345678",  
    "abc123"  
]
```

```
def check_password(password):
```

```
    if password.lower() in common_passwords:  
        print("\nPassword Strength: VERY WEAK")  
        print("Reason: This is a very common password.")  
        return
```

```
    score = 0
```

```
    if len(password) >= 8:  
        score += 1  
    else:  
        print("- Password should contain at least 8 characters.")
```

```
    if any(c.isupper() for c in password):  
        score += 1  
    else:  
        print("- Add at least one uppercase letter.")
```

```
    if any(c.islower() for c in password):  
        score += 1  
    else:  
        print("- Add at least one lowercase letter.")
```

```
    if any(c.isdigit() for c in password):  
        score += 1  
    else:  
        print("- Add at least one number.")
```

```
    if any(c in string.punctuation for c in password):
```

Security Check

Purpose

```
    score += 1
else:
    print("- Add at least one special character.")

if score <= 2:
    strength = "Weak"
elif score == 3 or score == 4:
    strength = "Medium"
else:
    strength = "Strong"

print("\nPassword Strength:", strength)

password = input("Enter your password: ")
check_password(password)
```

Example Test Results

```
*** Enter your password: 123456

Password Strength: VERY WEAK
Reason: This is a very common password.
```

```
*** Enter your password: hello1234
- Add at least one uppercase letter.
- Add at least one special character.

Password Strength: Medium
```

```
*** Enter your password: Hello123@

Password Strength: Strong
```

Part 2 – Port Scanning Using Nmap

Tool Used

- Kali Linux
- Nmap

Command Executed

nmap localhost

or

nmap 127.0.0.1

Scan Results

The Nmap scan was performed on the local machine (localhost).

Result:

Host is up.

All 1000 scanned TCP ports are closed.

No open TCP ports were detected during the scan.

Interpretation

The scan indicates that no network services were actively listening on the default TCP ports of the system.

This generally means:

- No unnecessary services are exposed.
- The attack surface of the machine is reduced.
- The system is less vulnerable to remote attacks through open ports.

Although no open ports were found, regular scanning remains important because newly installed applications or services may open ports in the future.

What If Ports Were Open?

Some common ports include:

Port	Service	Description
------	---------	-------------

22	SSH	Secure remote login
----	-----	---------------------

80	HTTP	Web server
----	------	------------

443	HTTPS	Secure web server
-----	-------	-------------------

21	FTP	File Transfer Protocol
----	-----	------------------------

3306	MySQL	Database service
------	-------	------------------

Open ports should only remain accessible if they are necessary for system operation.

Security Risks

Weak Passwords

Weak passwords are one of the leading causes of unauthorized access.

Attackers commonly use:

- Dictionary attacks
- Brute-force attacks
- Password spraying

Strong passwords significantly reduce the chances of compromise.

Open Ports

Open ports expose services to a network.

If these services:

- are outdated,
- misconfigured,
- or vulnerable,

they may allow attackers to gain unauthorized access.

Therefore, only required ports should remain open, and services should be updated regularly.

Nmap Output :

```
(abdul@kali)-[~]  
$ nmap 127.0.0.1  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-16 11:41 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000056s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.83 seconds
```

Reflection

This assignment helped me understand the importance of practical cybersecurity tools.

Developing a password strength checker demonstrated how simple programming techniques can encourage users to create stronger passwords.

Using Nmap showed how systems expose services through network ports and how administrators can identify potential security risks.

Although my scan did not reveal any open TCP ports, it reinforced the importance of regularly monitoring systems and ensuring unnecessary services remain disabled.

Overall, this exercise improved my understanding of password security, network reconnaissance, and basic defensive cybersecurity practices.

Conclusion

This project successfully introduced two important cybersecurity concepts:

- Password security
- Network port scanning

Strong passwords and careful monitoring of network services are essential components of protecting computer systems against cyber threats.